

安徽大学 2023—2024 学年第二学期

《高等数学 A (二)》期中考试试题参考答案及评分标准

一. 选择题 (每小题 3 分, 共 30 分)

1. D 2. D 3. A 4. B 5. A
6. C 7. A 8. C 9. D 10. B

二. 计算题 (每小题 11 分, 共 55 分)

11. 解: $\frac{\partial z}{\partial x} = f_1' \cdot 1 + f_2' \cdot \frac{1}{y} = f_1' + \frac{1}{y} f_2'$,
 $\frac{\partial^2 z}{\partial x \partial y} = f_{12}'' \cdot \left(-\frac{x}{y^2}\right) - \frac{1}{y^2} f_2' + \frac{1}{y} f_{22}'' \cdot \left(-\frac{x}{y^2}\right) = -\frac{x}{y^2} \left(f_{12}'' + \frac{1}{y} f_{22}''\right) - \frac{1}{y^2} f_2'$.
 (11 分)

12. 解: $\begin{cases} 2x + 2y \frac{dy}{dx} + 2z \frac{dz}{dx} = 0 \\ 1 + \frac{dy}{dx} + \frac{dz}{dx} = 0 \end{cases}$, 即 $\begin{cases} y \frac{dy}{dx} + z \frac{dz}{dx} = -x \\ \frac{dy}{dx} + \frac{dz}{dx} = -1 \end{cases}$, $J = \begin{vmatrix} y & z \\ 1 & 1 \end{vmatrix} = y - z \neq 0$,

解得 $\frac{dy}{dx} = \frac{-x \quad z}{-1 \quad 1} / J = \frac{z-x}{y-z}$, $\frac{dz}{dx} = \frac{y \quad -x}{1 \quad -1} / J = \frac{x-y}{y-z}$,

所以在点 $P(1, -2, 1)$ 处的切向量为 $\vec{T} = \{1, 0, -1\}$,

法平面方程为 $(x-1) - (z-1) = 0$, 即 $x - z = 0$.

注: 本题也允许其他解法。

..... (11 分)

13. 解: 令 $p = y'$, 则方程化为 $(1-2x) \frac{dp}{dx} = p$,

分离变量, 得 $\frac{dp}{p} = -\frac{1}{2x-1} dx$,

积分得 $\ln p = -\frac{1}{2} \ln(2x-1) + \ln C_1$, 或 $p = y' = C_1 (2x-1)^{-\frac{1}{2}}$,

再积分, 得原方程的通解为 $y = C_1 (2x-1)^{\frac{1}{2}} + C_2$.

..... (11 分)

14. 解: $\iint_D (x+y)^2 dx dy = \iint_D (x^2 + y^2 + 2xy) dx dy = \iint_D (x^2 + y^2) dx dy$
 $= \int_0^{2\pi} d\theta \int_0^1 r^2 \cdot r dr = \frac{\pi}{2}$

..... (11 分)

15. 解: $\iint_D |x+y-\pi| dx dy = \iint_{D_1} (\pi-x-y) dx dy + \iint_{D_2} (x+y-\pi) dx dy$

其中, $D_1: 0 \leq x \leq \pi, 0 \leq y \leq \pi, x+y \leq \pi$, $D_2: 0 \leq x \leq \pi, 0 \leq y \leq \pi, x+y > \pi$

$$\text{故原式} = \int_0^\pi dx \int_0^{\pi-x} (\pi - x - y) dy + \int_0^\pi dx \int_{\pi-x}^\pi (x + y - \pi) dy = \frac{\pi^3}{3}$$

..... (11 分)

三. 应用题 (共 10 分)

16. 解: 设长方体在第一卦限的顶点为 (x, y, z) . 该题为

$$V = 8xyz \text{ 在 } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \text{ 下的最大值.}$$

令 $F(x, y, z, \lambda) = xyz + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$. 解方程组

$$\begin{cases} \frac{\partial F}{\partial x} = yz + 2\lambda \cdot \frac{x}{a^2} = 0 \\ \frac{\partial F}{\partial y} = xz + 2\lambda \cdot \frac{y}{b^2} = 0 \\ \frac{\partial F}{\partial z} = xy + 2\lambda \cdot \frac{z}{c^2} = 0 \end{cases} \quad \text{解得 } x = \frac{a}{\sqrt{3}}, \quad y = \frac{b}{\sqrt{3}}, \quad z = \frac{c}{\sqrt{3}}.$$

由实际问题的意义知, 当边长为 $2x = \frac{2a}{\sqrt{3}}, 2y = \frac{2b}{\sqrt{3}}, 2z = \frac{2c}{\sqrt{3}}$

$$\text{有最大体积 } V = \frac{8abc}{3\sqrt{3}}.$$

..... (10 分)

四. 证明题 (共 5 分)

$$17. \text{ 证明: } f'_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{(\Delta x)^2 \cdot 0}{[(\Delta x)^2 + 0^2]^{3/2}} - 0}{\Delta x} = 0;$$

同理可得 $f'_y(0,0) = 0$, 即 $f(x, y)$ 在点 $(0,0)$ 处偏导数存在;

因为 $f'_y(0,0) = f'_x(0,0) = 0$, 所以

$$\Delta f(0,0) - [f'_x(0,0)\Delta x + f'_y(0,0)\Delta y] = \frac{(\Delta x)^2(\Delta y)^2}{[(\Delta x)^2 + (\Delta y)^2]^{3/2}} - 0 = \frac{(\Delta x)^2(\Delta y)^2}{[(\Delta x)^2 + (\Delta y)^2]^{3/2}},$$

考虑动点 (x, y) 沿着直线 $y = x$ 趋于 $(0,0)$, 那么 $\Delta x = \Delta y$, $\rho = \sqrt{2} |\Delta x|$, 此时

$$\frac{(\Delta x)^2(\Delta y)^2}{[(\Delta x)^2 + (\Delta y)^2]^{3/2}} = \frac{(\Delta x)^4}{4(\Delta x)^4} = \frac{1}{4} \neq 0, \text{ 即 } \lim_{\rho \rightarrow 0} \frac{\Delta f(0,0) - [f'_x(0,0)\Delta x + f'_y(0,0)\Delta y]}{\rho} \neq 0$$

故 $f(x, y)$ 在点 $(0,0)$ 处不可微.

..... (5 分)